Midterm 1: Worked Solutions

ECON 803 Fall 2026

Exam: October 16, 2026 Public release: October 18, 2026

This student edition matches the October 16, 2026, 90-minute exam. It includes all ten subparts and two optional optimization routes. It contains no student records, grading notes, or unreleased assessment solutions.

How to use this solution

First read your personalized PDF feedback on Canvas. Attempt a repair before looking at the corresponding worked solution; then close the solution and repeat the argument with one parameter or data row changed. Bring your feedback and the attempted problem to office hours, marking the first line you cannot justify.

A minimum reliable optimization answer. Write the choices, fixed parameters, domains and actual constraints. Compute the derivatives and state the full KKT conditions with a consistent sign convention. Solve one candidate; test its feasibility and any zero-choice entry condition. Recover the original choices, check the budget, and explain why the KKT certificate is globally sufficient. A negative interior candidate changes the active set, not the given parameters.

A minimum reliable revealed-preference answer. Identify which regime supplies the bundle and which regime supplies the comparison prices and income. Write the event implication first, then the probability inequality. Carry numerical counts through to shares, the correctly directed welfare interval, and a one-sentence conclusion. Independent cross-sections do not provide household pairings.

Start small. Reproduce one complete household regime and the two-regime numerical welfare calculation without the key. Then attempt an altered regime and one short maximum-comparison or event-inclusion proof. These are practice milestones, not a prediction of a core-exam result or a substitute for the full syllabus.

Reading map. Question 1 covers expenditure coordinates, KKT, human-search corners, agent pricing and finite-data maximum comparisons. Question 2 covers cross-affordability, welfare bounds, overlap and transitivity. The appendix offers optional shorter optimization routes; it is not an additional exam requirement.

Notation reminder. In Question 1, a is household search, A its productivity, b supplied AI search, F the fixed fee, $D = 1 + Aa + b$, $p = 1/D$, $e = pc$, c physical groceries and r outside consumption. In Question 2, r_{ts} is instead a probability: it is the share of regime-s bundles affordable at regime-t prices and income.

Question 1. AI as a tool and a hired shopping agent

(a) Expenditure coordinates and global optimality [6 points]

With $D = 1 + Aa + b > 0$, the price is $p = 1/D$, so $e = c/D > 0$ and $c = De$. This is a one-to-one change of variables for fixed A, b, F and a given a. Substitution gives

$$f(e, a, r; A, b) = -\frac{1}{De} - \frac{a}{8} + r, \qquad e + r \le 2 - F, \quad e > 0, \quad a, r \ge 0.$$

The feasible set is convex. The objective is concave in all choices and strictly concave in (e, a). Thus a feasible point with a valid KKT certificate is a global optimum. Budget exhaustion determines r; strict concavity then gives uniqueness of the full optimum.

(b) KKT conditions [10 points]

Use nonnegative multipliers λ, μ, η on $2 - F - e - r$, a, and r:

$$L = -\frac{1}{De} - \frac{a}{8} + r + \lambda(2 - F - e - r) + \mu a + \eta r.$$

Stationarity is

$$\frac{1}{De^2} - \lambda = 0, \qquad \frac{A}{D^2 e} - \frac{1}{8} + \mu = 0, \qquad 1 - \lambda + \eta = 0.$$

Primal and dual feasibility and complementary slackness are

$$e > 0, \quad a, r \ge 0, \quad e + r \le 2 - F, \qquad \lambda, \mu, \eta \ge 0,$$

$$\lambda(2 - F - e - r) = 0, \qquad \mu a = 0, \qquad \eta r = 0.$$

Equivalently the search entry surplus is $E_a = A/(D^2 e) - 1/8 \le 0$ with $aE_a = 0$. If $r > 0$, then $\eta = 0$, so $\lambda = 1$.

Question 1, continued

(c) Manual search, a tool, and delegation [14 points]

Here $F = 0$ and the student may assume $r > 0$. The budget binds, $\lambda = 1$, and grocery stationarity gives

$$e = D^{-1/2}, \qquad c = \sqrt{D}, \qquad r = 2 - D^{-1/2}.$$

Positive human effort requires

$$\frac{A}{D^{3/2}} = \frac{1}{8} \quad \Longrightarrow \quad D = (8A)^{2/3}.$$

At $a = 0$, instead check $A/(1+b)^{3/2} \leq 1/8$. The general solution with a positive numeraire is

$$D^* = \max\{(8A)^{2/3}, 1+b\}, \qquad a^* = \frac{D^* - 1 - b}{A}.$$

The three requested cases are:

Regime (A, b)	a	e	c	p	r	V
Manual $(1, 0)$	3	1/2	2	1/4	3/2	5/8
Tool $(8, 0)$	15/8	1/4	4	1/16	7/4	81/64
Delegated $(8, 63)$	0	1/8	8	1/64	15/8	7/4

Here $V = -1/c - a/8 + r$ is total household utility before a fee. For manual search and the tool, the interior search equality holds; the KKT multipliers are $(\lambda, \mu, \eta) = (1, 0, 0)$. For delegated service, positive human search would require $1 + 8a + 63 = 16$, giving $a = -6$, which is infeasible. At the actual boundary,

$$E_a = \frac{8}{64^{3/2}} - \frac{1}{8} = \frac{1}{64} - \frac{1}{8} = -\frac{7}{64} < 0.$$

Thus $(\lambda, \mu, \eta) = (1, 7/64, 0)$ certifies the corner. All budgets bind and all complementary products vanish. Concavity makes these unique global optima. The tool raises the productivity of human effort; delegated service supplies effective search even with zero human effort. Physical consumption rises while grocery expenditure falls. The cases are not claims about empirical AI effects.

Question 1, continued

(d) A profitable contract and consumer surplus [8 points]

The agent operates in the $A = 8$ environment. Let $W(b)$ be optimized household utility before the fee. The household's outside option is the powerful tool, $W(0) = 81/64$, not manual search. Its willingness to pay for delegation is $\Delta(b) = W(b) - W(0)$. Because the numeraire stays positive, a fixed fee reduces optimized utility one-for-one and does not change a, e, c.

Revenue is increasing in F, so the sharing restriction binds:

$$F(b) = \frac{1}{4}\Delta(b), \qquad \Pi(b) = \frac{1}{4}[W(b) - W(0)] - \frac{b^2}{7680}.$$

The supplied envelope identity gives $W'(b) = 1/(D^2 e)$. With $A = 8$ and $r > 0$, human search is positive when $b < 15$, then reaches zero at $b = 15$. Therefore

$$W(b) = \begin{cases} 81/64 + b/64, & 0 \le b \le 15, \\ 2 - 2/\sqrt{1+b}, & b \ge 15, \end{cases} \qquad W'(b) = \begin{cases} 1/64, & 0 \le b \le 15, \\ (1+b)^{-3/2}, & b \ge 15. \end{cases}$$

There is no derivative discontinuity at $b = 15$. For $b \le 15$, $\Pi'(b) = 1/256 - b/3840$, which vanishes at $b = 15$. Beyond 15, the marginal benefit decreases and the marginal cost increases, so $\Pi'(b) < 0$. Alternatively, the given concavity of W and strictly convex cost imply that Π is strictly concave: its stationary point is the unique global maximum. Thus

$$\boxed{b^* = 15, \quad F^* = \frac{15}{256}, \quad C(b^*) = \frac{15}{512}, \quad \Pi^* = \frac{15}{512} > 0.}$$

No service $(b, F) = (0, 0)$ yields zero profit and is dominated. At the optimal contract,

$$(a, e, c, p, r) = \left(0, \frac{1}{4}, 4, \frac{1}{16}, \frac{433}{256}\right), \qquad V = W(15) - F^* = \frac{369}{256}.$$

The tool's utility was $81/64 = 324/256$, so the consumer gains $45/256 > 0$. Yet outside consumption falls from $7/4 = 448/256$ to $433/256$: the fee exceeds the change in grocery expenditure, which is zero here. The welfare gain instead includes eliminating human effort $15/8$. A strict utility gain does not require increased cash.

Weak voluntary participation only imposes $F \le \Delta(b)$. A monopolist could set equality and leave zero consumer surplus. Here the explicit quarter-share cap guarantees a strictly positive consumer share whenever $\Delta(b) > 0$.

Question 1, continued

(e) Afriat-style inequalities from unique maximization [12 points]

This subpart uses a separate interior panel and the known price technology, as stated in the revised exam. At observation t, utility is strictly increasing in outside consumption, so the budget is exhausted:

$$r_t = M_t - p_t c_t.$$

Here M_t is income net of the date-specific fixed fee. For a comparison pair with $q_{jt}c_j \leq M_t$, define the counterfactual outside consumption $r_{jt} = M_t - q_{jt}c_j \geq 0$. Then (c_j, a_j, r_{jt}) is feasible at date t, with exactly the same AI assistance as the observed choice. By the definition of a maximum,

$$U_t + r_t \geq U_j + r_{jt}.$$

Substitute the two outside-consumption expressions and cancel income:

$$\begin{aligned} U_t + M_t - p_t c_t &\geq U_j + M_t - q_{jt}c_j, \\ U_j - U_t &\leq q_{jt}c_j - p_t c_t. \end{aligned}$$

If $(c_j, a_j) \neq (c_t, a_t)$, the counterfactual is different from the unique maximizer. The maximum comparison, and hence the final inequality, is strict. If the consumption-search pair is identical, both prices are evaluated at (A_t, b_t) and the inequality is equality, even when incomes or observed outside consumption differ across dates.

The correct counterfactual price is $q_{jt} = p(a_j; A_t, b_t)$: it combines the alternative's human search with date t's tool and contracted service. Using $p_j = p(a_j; A_j, b_j)$ would compare a different environment. If $q_{jt}c_j > M_t$, the alternative requires negative outside consumption and is not feasible; the displayed comparison is not imposed for that pair. No derivative, concavity, supporting plane, or multiplier is required in this subpart. These are necessary finite-data restrictions, not a sufficient rationalizability theorem.

Optional original-coordinate route. One may write the same maximum comparison directly as

$$u(c_t, a_t) + M_t - p(a_t; A_t, b_t)c_t \geq u(c_j, a_j) + M_t - p(a_j; A_t, b_t)c_j.$$

This is the same proof in original coordinates; counterfactual feasibility and strictness must still be checked.

Question 2. Population WARP

(a) Affordability and welfare [10 points]

On A_{ts}, $p^t \cdot x_\nu^s < m_t$ almost surely by the no-equality assumption. Choose a sufficiently small $\varepsilon > 0$ so that $x_\nu^s + \varepsilon \mathbf{1}$ is affordable at (p^t, m_t). Strictly increasing utility and optimality imply

$$V_\nu^t \geq u_\nu(x_\nu^s + \varepsilon \mathbf{1}) > u_\nu(x_\nu^s) = V_\nu^s.$$

Thus $A_{ts} \subseteq H_{ts}$ almost surely. Taking probabilities gives $r_{ts} \leq \theta_{ts}$. A bundle strictly inside the other regime's budget can be improved there, so that regime delivers strictly greater welfare.

(b) Welfare bounds and cross-section identification [12 points]

Reversing regimes gives $A_{st} \subseteq H_{st}$. The strict welfare events H_{ts} and H_{st} are disjoint, including when some types are indifferent. Therefore

$$r_{st} \leq \Pr(H_{st}) \leq 1 - \Pr(H_{ts}) = 1 - \theta_{ts}.$$

Combining with part (a),

$$\boxed{r_{ts} \leq \theta_{ts} \leq 1 - r_{st}}, \qquad \boxed{r_{ts} + r_{st} \leq 1}.$$

The economist computes r_{ts} by pricing each observed regime-s bundle at p^t and comparing with m_t. This uses only the marginal distribution of regime-s choices. Reverse the calculation using the regime-t sample for r_{st}. No household pairing is needed. A stable type distribution ensures both shares refer to the same latent population; without stability, their sum need not constrain a common population's behavior.

(c) Numerical welfare interval [10 points]

Five of the ten regime-2 cross-expenditures are below $m_1 = 12$, so $r_{12} = 5/10 = 0.50$. Four of the ten regime-1 cross-expenditures are below $m_2 = 13$, so $r_{21} = 4/10 = 0.40$. Thus

$$r_{12} + r_{21} = 0.90 \leq 1, \qquad \boxed{\theta_{12} \in [0.50, 0.60]}.$$

The pair satisfies population WARP with slack 0.10. Under the model, between 50% and 60% are better off in regime 1. The calculation is not a sampling-uncertainty test: the prompt treats the proportions as exact.

Question 2, continued

(d) A forced share of individual WARP violations [10 points]

Let $A = A_{12}$ and $B = A_{21}$ for the alternative population. Budget exhaustion remains maintained; this calculation relaxes universal individual WARP, not budget exhaustion. Then

$$\Pr(A \cap B) \geq \Pr(A) + \Pr(B) - 1 = 0.64 + 0.44 - 1 = 0.08.$$

On the intersection, each regime's chosen bundle is strictly affordable under the other budget almost surely. The choices are distinct, since identical bundles would have cross-expenditure equality by budget exhaustion. Mutual strict affordability violates individual WARP. Therefore

$$\boxed{\text{at least } 8\% \text{ of households must violate individual WARP.}}$$

Consequently, universal WARP is incompatible with these exact shares. The unknown overlap prevents the identification of particular households, and the total violating share could exceed the bound.

(e) Transitivity across three regimes [8 points]

Return to stable utility maximization. For every type,

$$H_{12} \cap H_{23} \subseteq H_{13},$$

because $V_\nu^1 > V_\nu^2$ and $V_\nu^2 > V_\nu^3$ imply $V_\nu^1 > V_\nu^3$. Hence

$$\begin{aligned} \theta_{13} &\geq \Pr(H_{12} \cap H_{23}) \\ &\geq \Pr(H_{12}) + \Pr(H_{23}) - 1 \\ &\geq r_{12} + r_{23} - 1 = 0.72 + 0.65 - 1 = 0.37. \end{aligned}$$

The direct bound is $\theta_{13} \geq r_{13} = 0.20$, so

$$\boxed{\theta_{13} \geq \max\{0.20, 0.37\} = 0.37.}$$

At least 37% are better off in regime 1 than in regime 3. This extension uses transitive utility comparisons in the maintained rational model. Pairwise WARP alone is not asserted to imply transitivity.

Optional appendix: shorter economic KKT routes

Method 1: numeraire normalization and entry screening. With a positive numeraire, set $\lambda = 1$ and use only

$$e = D^{-1/2}, \qquad E_a = AD^{-3/2} - \frac{1}{8} \leq 0, \qquad aE_a = 0.$$

At $a = 0$, $D = 1 + b$. Entry has positive surplus if $1 + b < (8A)^{2/3}$, so that boundary is rejected. Otherwise the boundary passes. A positive-search candidate requires $D = (8A)^{2/3}$. Recover $\mu = -E_a$ and $\eta = 0$. At the threshold the entry surplus is zero, not strictly negative: uniqueness follows from the global certificate. This method is the shortest route to the comparative table; no full second KKT calculation is needed for each case.

Method 2: budget reduction and one-dimensional optimization. Set $r = 2 - F - e$, retaining $0 < e \leq 2 - F$. Write $d = 1 + b$ and $a = (D - d)/A$, so $D \geq d$. Maximization is equivalent to minimizing

$$J(e, D) = e + \frac{1}{De} + \frac{D - d}{8A}.$$

For a fixed D, the sharp inequality

$$e + \frac{1}{De} \geq \frac{2}{\sqrt{D}}, \qquad e + \frac{1}{De} - \frac{2}{\sqrt{D}} = \frac{(e - D^{-1/2})^2}{e}$$

gives the unique minimizing expenditure. It must satisfy the retained wealth constraint, which holds in (c) and for the contracts in (d). The remaining derivative is

$$\frac{d}{dD}\left(\frac{2}{\sqrt{D}} + \frac{D - d}{8A}\right) = \frac{1}{8A} - D^{-3/2}.$$

Thus $D^* = \max\{(8A)^{2/3}, d\}$. Recover a, e, c, p, r. This route is global. The explicit KKT system is still required in (b), and a zero-effort check is still required in (c).

Verification scope. The worked derivations and exact rational calculations have been checked against the distributed exam. No Lean verification is claimed. The released LyX file is the editable source; the PDF is exported from it. These classroom AI regimes are illustrative counterfactuals, not empirical estimates. The finite-data comparisons are necessary restrictions, not a complete rationalizability theorem.